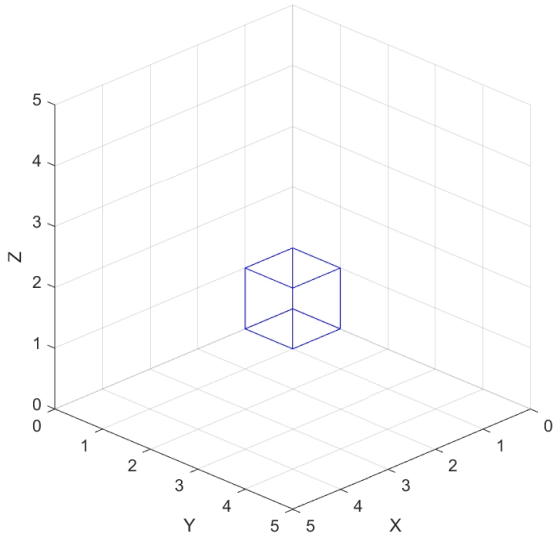
Lab 2 – Coordinate Transforms 3D

1 Introduction

In this lab the mathematics behind 3D coordinate systems and transformation between coordinate frames will be exercised. MATLAB will be used initially to become familiar with the operations and setup of coordinate transforms.

2 3D Translation and Rotation



1) Assign a coordinate system {C} to coordinate (0,0,0) of a cube with each side equal to 1. Place the cube at the origin of the world coordinate system {W} with no rotation and plot.

2) Using a FOR loop for animation, translate {C} to (2,2,2) and rotate 360 degrees about the Z axis in {C}.

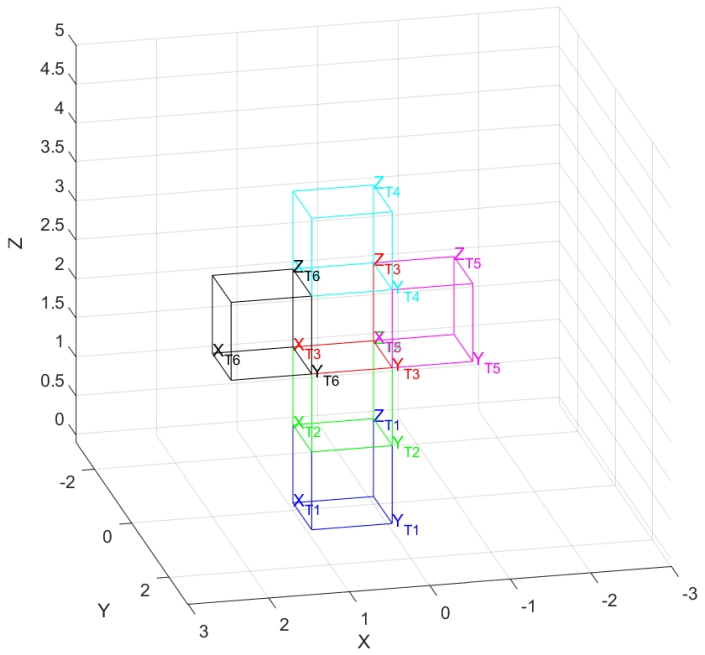
3) Using a FOR loop for animation, translate {C} to (2,2,2) and rotate 360 degrees about the Z axis in {W}.

The sample code in the 2D case can be used with the addition of plotting in a 3D volume with 3D points instead. The *view* command sets the camera view azimuth and elevation.

```
plotvol([0 5 0 5 0 5]);
view(115,25);
plot3 (pts(:,1),pts(:,2),pts(:,3),'b-');
```

3 3D Robot

1) Stack a vertical set of 4 cubes and a horizontal set of 3 cube across at the 3rd cube from the top and plot the resulting robot. All cubs have sides of length of 1. For each cube assign a coordinate system with the cube at (0,0,0) and then determine the transformation chain to get back to the world coordinate system to convert from the point in T? to W.



3) Using a FOR loop for animation, translate the robot 360 degrees about the Z axis in {W}.